

Sam Armstrong

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PERSONAL PROFILE

I am currently completing my Master's degree in Computer Science, after 3+ years of industry experience building software that is maintainable, scalable, and genuinely useful for customers. I am driven by developing, shipping and iterating rapidly, and am seeking a company culture that shares this same philosophy.

EDUCATION

2025–2026

University of Warwick

MSc

Computer Science

2018–2021

University of Sheffield

BSc

Artificial Intelligence & Computer Science

First Class Honours

EXPERIENCE & PROJECTS

Hackathons

- Whack (2025) — Created an AI learning tool that lectures over pdf slides, asks questions, and learns the user's level of understanding for the incident.io problem statement.
- Anthropic London (2023) — Created a GitHub bot to automatically add docstrings to functions using the Claude API.

Presented My Academic Research

- Workshop on Multi-Agent Based Simulation (MABS) 2022
- Sheffield SIAM-IMA Applied Mathematics and Statistics Conference 2021

ENTREPRENEURSHIP

2025–Present

Founder — CozyCasts

Created CozyCasts — a platform for streaming AI-generated podcasts to help you learn, or for entertainment. The backend is designed for scalability, while maintaining low latency and cost-conscience, and uses an API/worker pattern to decouple the compute-heavy TTS from the API.

- API service is deployed serverless on Google Cloud Run
- Worker is deployed on GKE with KEDA for scale-to-zero
- API service schedules worker jobs using Google Cloud Pub/Sub
- Redis Pub/Sub is used for streaming audio back to the API as it's generated in real-time, then on to the user via a WebSocket
- We use cache-aside to prevent a race condition on the WebSocket
- Metadata is stored in PostgreSQL (Cloud SQL) and audio/images in GCS
- Frontend webapp is built using React/Next.js

CozyCasts is currently in alpha, available at cozycasts.com

2024–2025

Co-Founder & CTO — Ivy

Co-founded Ivy — which allows any ML models or code to be converted between frameworks such as PyTorch and JAX — as a spin-off from our previous company, Unify. Primarily working with:

- DevOps with cloud infrastructure (GCP) and CI/CD
- Advanced Python programming; AST; ML frameworks

I continue to maintain Ivy now as an open-source project, which is:

- Available via PyPI ('pip install ivy') and has ~14,000 GH stars
- Integrated into Kornia, the popular computer vision library

EMPLOYMENT

2023–2024

Machine Learning Engineer — Unify AI (YC W23)

- Led a team of five developing a computational graph tracer which can extract a minimal functional graph from arbitrary Python code — enabling any Ivy model to be compiled into an efficient representation.
- Developed the multi-backend framework Ivy: creating backend functions, forming the CI, and reviewing PRs from contributors.

2022–2023

ML Software Engineer — Intradiem

- Orchestrated the creation of Java data pipelines for transforming raw data from an AWS S3 datalake into bucketed time-series data. These were used for both ML applications in both training and inference, and required performant algorithm design for scalability.
- Led development of a Python micro-service using Flask/MariaDB, containerised with Docker, to document customer/product relationships and make this information easily accessible to services across the business.